

Description

BP2636D is a Boost PFC driver with high efficiency, high PF and low THD. The device operates in critical conduction mode and is better for EMI and Efficiency improvement.

BP2636D utilizes MOSFET gate driving technique without any auxiliary winding. With very few external components count, it can achieve excellent constant voltage performance, so as to reduce the system cost and size greatly.

BP2636D offers rich protection functions to improve the system reliability, including load open circuit protection (Over Voltage Protection), MOSFET over current limit and over temperature protection.

BP2636D is available in SOP-8 package and the thermal condition is improved by optimized pin configuration.

Features

- PF>0.9, THD<10% at Universal Input
- Single-winding inductor for simple design
- Internal 500V MOSFET
- Critical Conduction Mode
- HV JFET for fast startup
- Voltage reference accuracy of up to $\pm 2\%$
- Integrated protections
 - Output Over Voltage Protection
 - MOSFET Over Current Protection
 - VCC VULO Protection
 - Over Temperature Protection
- Available in SOP-8 Package

Applications

- BOOST APFC pre-converter



SOP-8 Package

Typical Application

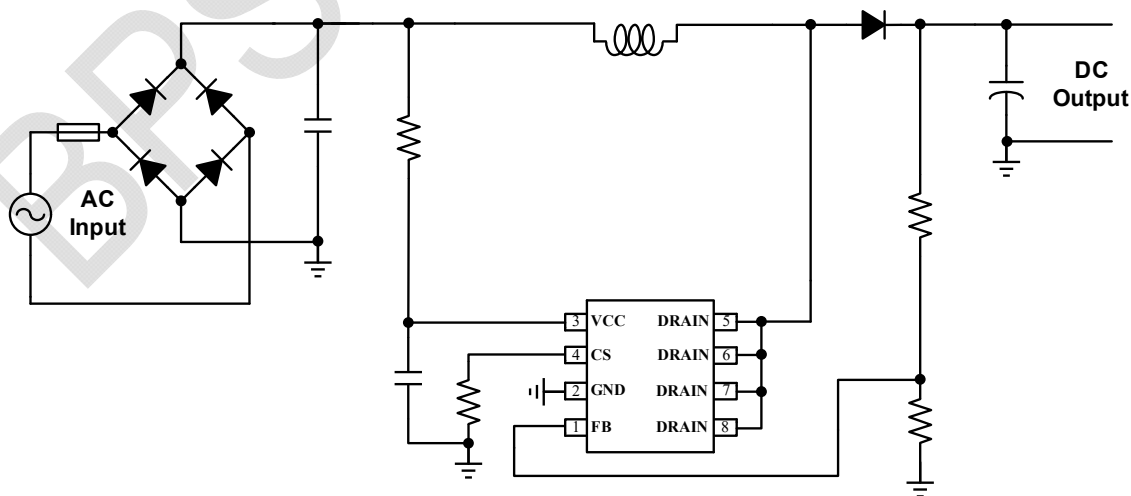


Figure 1 BP2636D Typical Application

Ordering Information

Part Number	Package	Package Method	Marking
BP2636D	SOP-8	Tape 4,000/Reel	BP2636 XXXXXYZ XXYYWWD

Pin Configuration and Marking Information

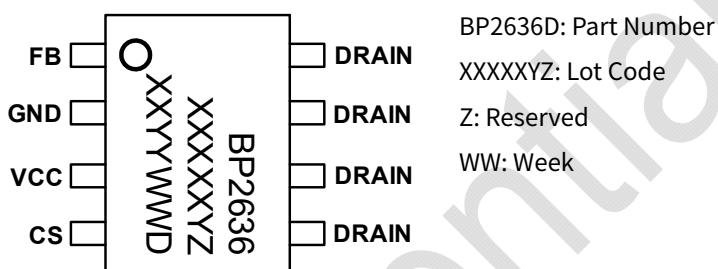


Figure 2 Pin Configuration

Pin Definition

Pin No.	Name	Description
1	FB	Boost output voltage sensing and feedback
2	GND	IC Ground
3	VCC	IC Power Supply
4	CS	Boost MOSFET Current Sensing
5, 6, 7, 8	DRAIN	DRAIN of Internal MOSFET

Absolute Maximum Ratings (注 1)

Symbol	Parameters	Range	Units
VCC	IC power supply	-0.3~20	V
V _{CC_CLAMP}	VCC internal clamp voltage	15	V
CS	MOSFET current sensing	-0.3~6	V
FB	Output voltage sensing and feedback	-0.3~6	V
DRAIN	DRAIN of Internal MOSFET	-0.3~500	V
P _{DMAX}	Power dissipation (Note 2)	0.45	W
θ_{JA}	Thermal resistance of junction to ambient (Note 3)	145	°C/W
T _J	Operating junction temperature	-40~150	°C
T _{STG}	Storage temperature range	-55~150	°C
ESD	HBM ESD (Note 4)	2	kV

Note 1: Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. Under “recommended operating conditions” the device operation is assured, but some particular parameter may not be achieved. The electrical characteristics table defines the operation range of the device, the electrical characteristics is assured on DC and AC voltage by test program. For the parameters without minimum and maximum value in the EC table, the typical value defines the operation range, the accuracy is not guaranteed by spec.

Note 2: The maximum power dissipation decrease if temperature rise, it is decided by T_{JMAX}, θ_{JA} , and environment temperature (T_A). The maximum power dissipation is the lower one between $P_{DMAX} = (T_{JMAX} - T_A) / \theta_{JA}$ and the number listed in the maximum table.

Note 3: Tested on 1 in.² double-sided board according to JEDEC standard.

Note 4: Tested According to JEDEC standard, 100pF capacitor discharge on 1.5KΩ resistor.

Electrical Characteristics (Note 5) (Unless otherwise specified, $T_A = 25^\circ\text{C}$)

Symbol	Description	Condition	Min	Typ	Max	Units
Supply Voltage Section						
V_{CC_CLAMP}	V_{CC} Clamp Voltage	$I_{CC}=1\text{mA}$	14	15	16	V
V_{CC_ON}	V_{CC} turn on threshold	V_{CC} rising	11	12.5	14	V
V_{CC_UVLO}	V_{CC} UVLO threshold	V_{CC} falling	7	8	9	V
I_{OP}	Omniscient current	No switching	285	380	475	μA
MOSFET Over Current Protection						
V_{CS_LIM}	CS OCP threshold		450	500	550	mV
Output Voltage Regulation						
V_{FB_REF}	FB reference voltage		2.45	2.5	2.55	V
V_{OVP_REF}	OVP threshold		2.64	2.7	2.76	V
V_{OVP_REL}	OVP release threshold		2.5	2.575	2.65	V
Internal Time Control						
T_{OFF_MIN}	Minimum off time			3.3		μs
T_{OFF_MAX1}	Maximum off time 1	$V_{FB} \leq 1.67\text{V}$		25		μs
T_{OFF_MAX2}	Maximum off time 2	$V_{FB} > 1.67\text{V}$	35	50	65	μs
T_{LEB}	LEB time			350		ns
T_{ON_MAX}	Max on time		30	35	40	μs
$T_{DET_BLANKING}$	ZCD blanking time			1.3		μs
Power MOSFET						
R_{DS_ON}	R_{DS_ON} of internal MOSFET	$V_{GS}=10\text{V}/I_{DS}=2.0\text{A}$		2.1		Ω
BV_{DSS}	Breakdown voltage of internal MOSFET	$V_{GS}=0\text{V}/I_{DS}=250\mu\text{A}$	500			V
Thermal Shutdown						
T_{OTP}	OTP threshold			160		$^\circ\text{C}$
T_{OTP_HYS}	OTP hysteresis			15		$^\circ\text{C}$

Note 5: The maximum and minimum parameters specified are guaranteed by test, the typical value are guaranteed by design, characterization and statistical analysis.

Internal Block Diagram

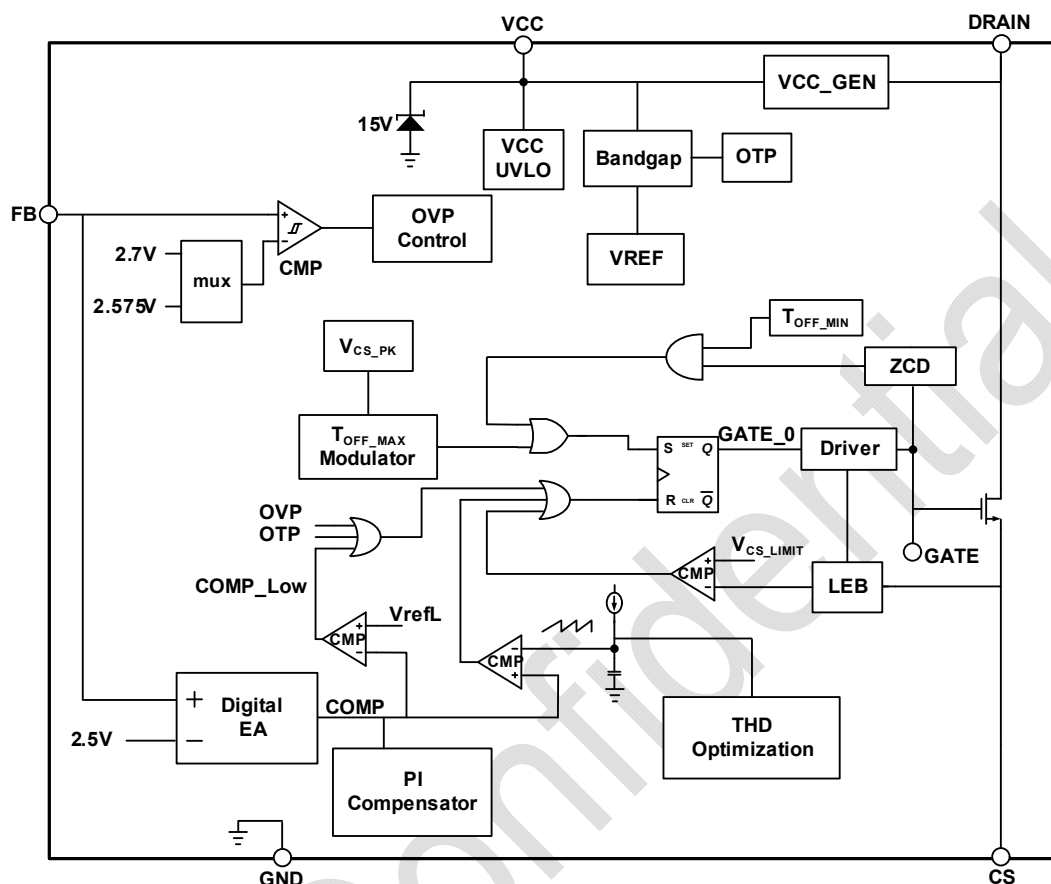


Figure 3 BP2636D Internal Block Diagram

Application Information

BP2636D is a Boost PFC driver with high efficiency, high PF and low THD. The device operates in critical conduction mode and is better for EMI and Efficiency improvement.

BP2636D utilizes MOSFET gate driving technique without any auxiliary winding. With very few external components count, it can achieve excellent constant voltage performance, so as to reduce the system cost and size greatly.

Start-up

After system is powered up, the VCC pin capacitor is charged up by internal HV JFET. After VCC voltage is

higher than turn on threshold, the internal circuits start operating. BP2636D integrates a 15V Clamper inside for clamping VCC voltage. In order to reduce the power dissipation of the device, additional resistors can be connected to the rectified input voltage, to reduce the supply current through JFET.

Out Voltage Regulation & OVP Setting

Output voltage is sensed by FB pin and ON time of MOSFET is regulated to keep FB voltage at 2.5V as a closed loop control. Output Voltage Protection (OVP) is also implemented by FB pin. When FB voltage reaches 2.7V, OVP is triggered. See Figure 4 below for the wiring

diagram of FB pin.

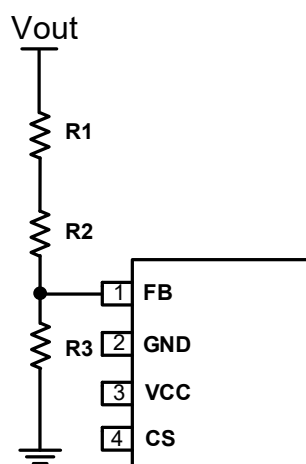


Figure 4 FB Wiring Diagram

Regulated output voltage and OVP voltage is set as:

$$V_{OUT} = \frac{R_1 + R_2 + R_3}{R_3} \times 2.5V$$

$$V_{OVP} = \frac{R_1 + R_2 + R_3}{R_3} \times 2.7V$$

Where:

R_3 is the resistance of lower resistor;

R_1 and R_2 is the resistance of upper resistor;

V_{OUT} is the output voltage;

V_{OVP} is output OVP voltage;

In order to improve system efficiency, FB lower resistance can be sized to about 5~10KΩ. A filter capacitor can be connected to FB for better interference immunity.

Over Current Protection

Cycle by cycle current sense is adopted in BP2636D, the CS pin is connected to the current sense comparator, and the voltage on CS pin is compared with the internal reference voltage. The MOSFET will be switched off when the voltage on CS pin reaches the threshold.

The peak inductor current is given by:

$$I_{PK_LMT} = \frac{500}{R_{CS}} (mA)$$

Where: R_{CS} is the current sense resistor value.

CS comparator includes a leading edge blanking time of 300ns.

Boost Inductor

BP2636D works under inductor current critical conduction mode. When the power MOSFET is switched on, the current of the inductor rises up from zero, the on time of the MOSFET can be calculated by the equation:

$$t_{on} = \frac{L \times I_{PK}}{V_{IN}}$$

Where: L is the inductance;

I_{PK} is the peak current of inductor;

V_{IN} is rectified input voltage;

Max. on time of the device is 35us.

After the power MOSFET is switched off, the current in the inductor decrease. When the inductor current reaches zero, the power MOSFET is turned on again by internal logic of the device. The OFF time of the MOSFET is given by:

$$t_{off} = \frac{L \times I_{PK}}{V_{OUT} - V_{IN}}$$

Boost inductance is set as:

$$L = \frac{(V_{OUT} - V_{IN}) \times V_{IN}}{f \times I_{PK} \times V_{OUT}}$$

Where:

f is the minimum switching frequency Switching frequency of BP2636D is related to input voltage. Usually, minimum switching frequency occurs at peak voltage over one cycle at low line voltage.

Protections

BP2636D offers rich protection functions, including Output Voltage Protection (OVP), VCC UVLO, MOSFET cycle-by-cycle Over Current Protection (OCP), Over Temperature Protection (OTP) and FB Short Protection..

Output Voltage Protection

When the load is open, FB pin voltage will rise with output voltage. When the FB pin voltage reaches 2.7V, the device will trigger Over Voltage Protection and stop switching. When FB pin voltage is lower than 2.575V, the system will recover to normal operation.

MOSFET Over Current Protection

The peak current of the inductor will rise as input voltage decreases. At lower input voltage, inductor peak current is higher. Fine tuning CS resistance is useful to limit inductor current.

Over Temperature Protection

When the junction temperature is over 160°C, the system will trigger the Over Temperature Protection and stop switching. When the junction temperature is lower than 145 °C, the system will recover to normal operation.

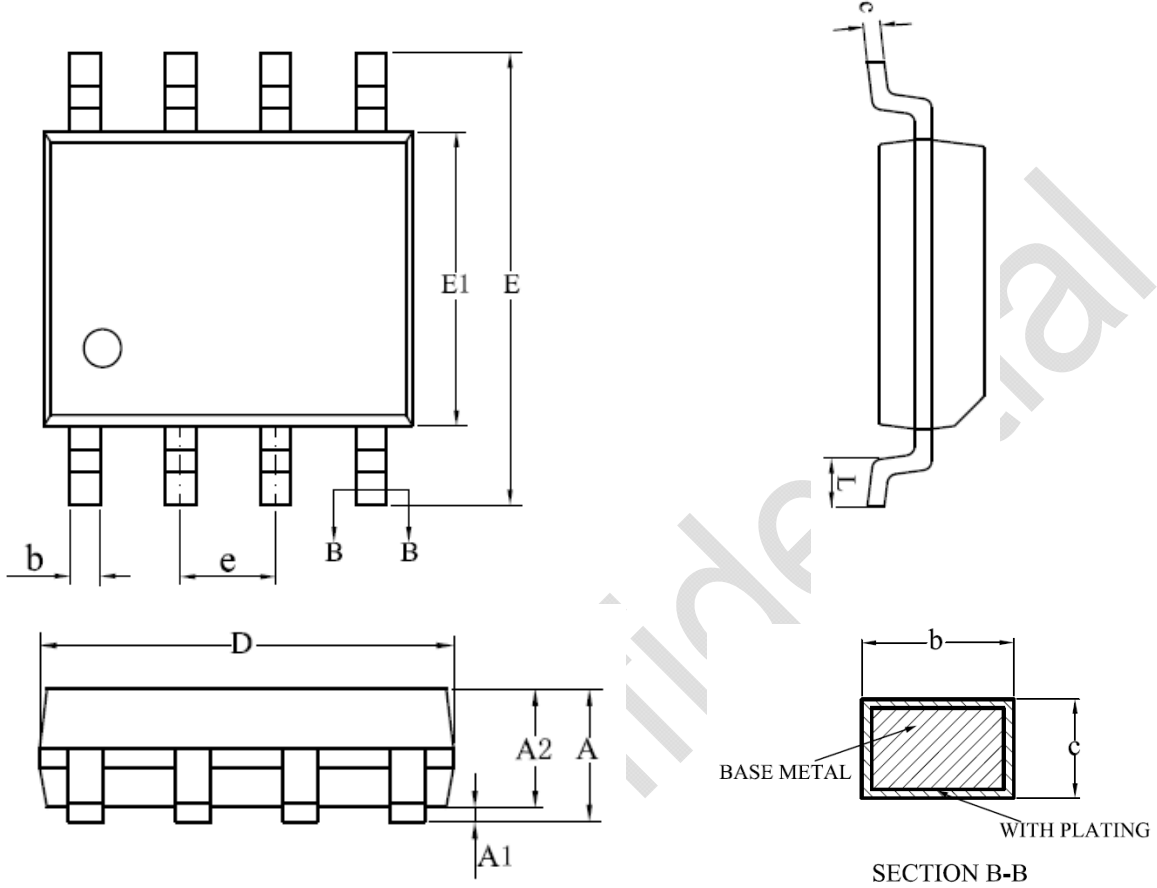
PCB Layouts

The following rules should be followed in BP2636D PCB layout:

- 1) The bypass capacitor on VCC pin should be as close as possible to the VCC and GND pin.
- 2) FB pin resistor divider should placed as close as possible to the FB pin, and the trace must be kept away from drain of MOSFET and far from noise source.
- 3) Ground of current sense resistor should be short and wide, and it should be as close as possible to the IC GND pin. Furthermore, ground of VCC, FB and COMP resistors as well as the capacitors should be connected to IC GND pin with short traces and should be separated from power ground path.
- 4) CS pin and current sensing resistor should be as close as possible and away from the high voltage switching nodes.
- 5) The area of main current loop should be as small as possible to reduce EMI radiation, such as the power MOSFET, the freewheeling diode, the inductor and the bus capacitor.

Package

SOP-8 PACKAGE OUTLINE DIMENSIONS



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	1.30	—	1.80
A1	0.05	—	0.25
A2	1.25	1.40	1.65
b	0.33	—	0.51
c	0.17	—	0.25
D	4.70	4.90	5.10
E	5.80	6.00	6.20
E1	3.70	3.90	4.10
e	1.27BSC		
L	0.40	—	1.00

Revision Information

Revision	Date	Notes
Rev. 1.0	2021/01	First Issue

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